

2022 Year 11 Examination

Mathematics Standard

Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	B
2	C
3	D
4	A
5	B
6	A
7	B
8	C
9	B
10	D
11	D
12	A
13	B
14	A
15	D

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Section II

Question 16

16 (a) (i)

Criteria	Marks
• Correct solution with no errors	2
• Some correct working	1

Sample answer

$$(2x + y)^3 = 8$$

$$2x + y = 2$$

$$y = 2 - 2x$$

16 (b) (i)

Criteria	Marks
• Correct solution with no errors	2
• Some correct working	1

Sample answer

$$\begin{aligned}BAC_{Female} &= \frac{10 \times 4 - 7.5 \times 4}{5.5 \times 75} \\&= 0.04848 \dots \\&= 0.05\end{aligned}$$

16 (b) (ii)

Criteria	Marks
• Correct solution with no errors • Must show evidence to support final answer	2
• Some correct working	1

Sample answer

$$\begin{aligned}time &= \frac{0.05}{0.015} \\&= 3.333333 \dots \\&= 3h \ 20min\end{aligned}$$

No, Sarah will not be able to leave by midnight as her BAC will be zero at 1:20am the following day.

16 (b) (iii)

Criteria	Marks
• List two limitations that affect BAC	2
• List one limitation that affects BAC	1

Sample answer

Many factors can affect Sarah's BAC that are not accounted for in this formula:

- The amount of food consumed prior to or throughout the night
- Body fat and muscle
- Alcohol tolerance
- Medication she may be taking
- Fatigue, stress mood

16 (c) (i)

Criteria	Mark
• Correct solution	1

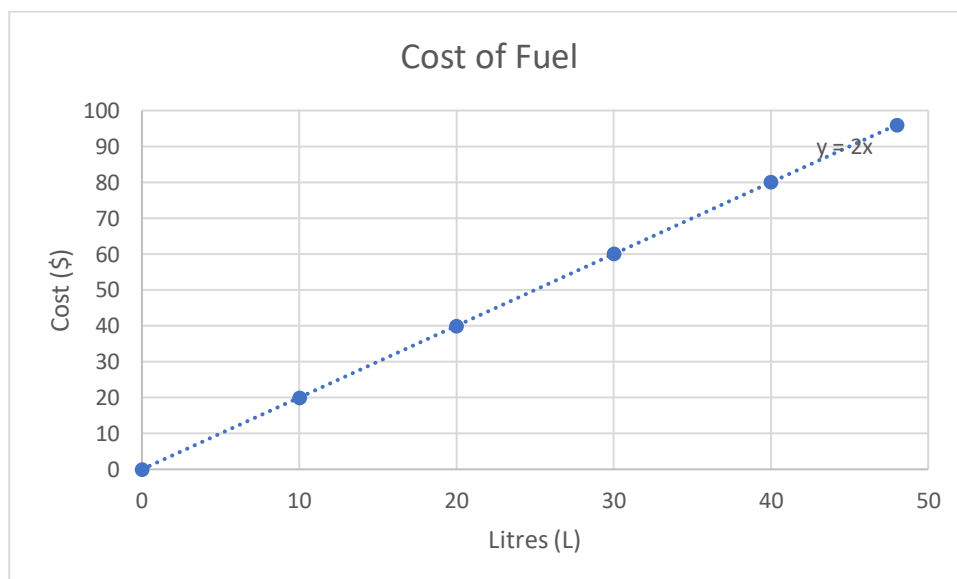
Sample answer

$$48 \times \$2.00 = \$96$$

16 (c) (ii)

Criteria	Marks
• Correct graph, including labelling of x - and y -axes; correct scale and intervals; straight line and labelling of graph	2
• One or more elements missing from the graph	1

Sample answer



16 (c) (iii)

Criteria	Marks
• State the gradient AND its significance	2
• State the gradient OR its significance	1

Sample answer

The gradient is 2. It is the rate of cost per litre, i.e. \$2.00/L.

16 (c) (iv)

Criteria	Mark
• State one limitation	1

Sample answer

Answers may vary and can include:

- Negative values for either litres or cost cannot be possible and therefore cannot be graphed
- Cannot go past 48 litres as this is the maximum fuel tank capacity of the car

Question 17

17 (a) (i)

Criteria	Mark
• Correct solution	1

Sample answer

$$\frac{0.1}{2} = 0.05 \text{ cm}$$

17 (a) (ii)

Criteria	Marks
• Correctly calculates the upper bound AND the lower bound using the answer in part (i) to find the limits of accuracy of the length of the radius	2
• Correctly calculates either the upper bound OR the lower bound using the answer in part (i) to find the limits of accuracy of the length of the radius	1

Sample answer

$$9.6 + 0.05 = 9.65 \text{ cm}$$

$$9.6 - 0.05 = 9.55 \text{ cm}$$

Therefore, the limits of accuracy are greater than or equal to 9.55cm but less than 9.65cm.

17 (a) (iii)

Criteria	Mark
Correctly covert the volume to capacity	1

Sample answer

$$3474 \text{ cm}^3 = 3464 \text{ mL}$$

$$= 3.464 \text{ L}$$

17 (a) (iv)

Criteria	Marks
Correctly calculates the upper bound AND the lower bound using the answer in part (ii) to find the limits of accuracy of the volume	2
Correctly calculates the upper bound OR the lower bound using the answer in part (ii) to find the limits of accuracy of the volume	1

Sample answer

$V = \pi r^2 h$ $= \pi \times 9.55^2 \times 12$ $= 3438 \text{ cm}^3$ (to the nearest whole)	$V = \pi r^2 h$ $= \pi \times 9.65^2 \times 12$ $= 3511 \text{ cm}^3$ (to the nearest whole)
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Therefore, the true volume of the cylinder is greater than or equal to 3438 cm^3 but less than 3511 cm^3 .

17 (b) (i)

Criteria	Marks
- Correctly calculate unknown side using Pythagoras' Theorem - Correctly add all sides to find the perimeter	3
One error in calculations	2
Multiple errors in calculations	1

Sample answer

$$c^2 = 6^2 + 8^2$$

$$c^2 = 100$$

$$c = 10$$

$$\begin{aligned}\therefore P &= \frac{2\pi(4)}{2} + 6 + 10 \\ &= 29 \text{ m (to the nearest metre)}\end{aligned}$$

17 (b) (ii)

Criteria	Marks
- Correctly calculate the total area of the land - Correctly calculate the number of pavers needed, and hence the total cost	3
Some correct working	2
More than two errors in working	1

Sample answer

$$\begin{aligned}A &= \frac{1}{2} \times \pi \times 4^2 + \frac{1}{2} \times 6 \times 8 \\ &= 49.13274123 \dots\end{aligned}$$

$$49.13274123 \div 0.5 = 98.26548 \dots$$

$\therefore$ 99 pavers needed.

$$\therefore \$12 \times 99 = \$1188$$

17 (c)

Criteria	Marks
Correctly calculate the total area using TWO applications of the trapezoidal rule	3
Mostly correct working, using TWO applications of the trapezoidal rule to calculate the total area	2
Multiple errors in working	1

Sample answer

$$\begin{aligned}Area_1 &= \frac{20}{2} (0 + 22) \\ &= 220 \text{ m}^2\end{aligned}$$

$$\begin{aligned}Area_2 &= \frac{20}{2} (22 + 15) \\ &= 370 \text{ m}^2\end{aligned}$$

$$\begin{aligned}Total \text{ area} &= 220 + 370 \\ &= 590 \text{ m}^2\end{aligned}$$

17 (d)

Criteria	Marks
• Correctly calculate the total amount of energy consumed	3
• Mostly correct working	2
• Some correct working	1

Sample answer

Chicken Stitzel: $525 \times 2 = 1050 \text{ kJ}$

Hot chips: $\frac{400}{250} \times 1830 = 2928 \text{ kJ}$

Watermelon: $3 \times 410 = 1230 \text{ kJ}$

Total consumed: $1050 + 2928 + 1230 = 5208 \text{ kJ}$

17 (e) (i)

Criteria	Mark
• Correctly states the number of days in the supply period	1

Sample answer

91 days.

17 (e) (ii)

Criteria	Mark
• Correctly states the total amount of electricity used	1

Sample answer

$609 + 729 = 1338 \text{ kWh}$

17 (e) (iii)

Criteria	Mark
• Correctly states the percentage of usage in the off-peak period.	1

Sample answer

$\frac{729}{1338} \times 100 = 54.5\% \text{ (to one decimal place)}$

17 (e) (iv)

Criteria	Mark
• Correctly calculates the value of A	1

Sample answer

$$224 \text{ kWh} \times \$0.3076 = \$68.90$$

Question 18

18 (a)

Criteria	Marks
• Correctly calculates the time in Sydney, accommodating for daylight savings, and shows full working out	2
• Some correct working	1

Sample answer

If it is 4:30pm in Queensland when the call begins, it is 5:30pm in Sydney.

$\therefore 5:30\text{pm} + 45\text{min} = 6:15\text{pm}$, Sydney time, when the call finishes.

18 (b)

Criteria	Marks
• Correctly calculates the arrival time of all THREE cities from Sydney	4
• Provides a concluding statement to support calculations	
• One error in calculations	3
• Two errors in calculations	2
• Three errors in calculations	1

Sample answer

Sydney to Madrid:

Sydney (12:00am Jan 1) – 9 hours = 3:00pm Dec 31 in Madrid when plane departs Sydney.

3:00pm + 24 hours in travel time = 3:00am Jan 1 arrival in Madrid.

$\therefore$ *Not an option.*

Sydney to Chile:

Sydney (12:00am Jan 1) – 14 hours = 10:00am Dec 31 in Chile when plane departs Sydney.

10:00am + 14.5 hours in travel time = 12:30am Jan 1 arrival in Chile.

$\therefore$ *Not an option.*

Sydney to Los Angeles:

Sydney (12:00am Jan 1) – 17 hours = 7:00am Dec 31 in Los Angeles when plane departs Sydney.

7:00am + 12 hours in travel time = 7:00pm Dec 31 arrival in Los Angeles.

$\therefore$ *Los Angeles is the second city that can be visited in time to celebrate New Years' Eve.*

18 (c) (i)

Criteria	Mark
• Correct answer	1

Sample answer

27 minutes.

18 (c) (ii)

Criteria	Mark
• Correct answer	1

Sample answer

06:41

18 (c) (iii)

Criteria	Marks
• Correct solution showing working out to support answer	2
• One error made	1

Sample answer

8 minutes – 2 minutes = A total of 6 minutes late.

Since it was the 06:39 lightrail that was late, if on time it would have arrived at 06:57.

It was 6 minutes late and therefore arrived at 07:03 instead.

Question 19

19 (a) (i)

Criteria	Marks
• Correct solution with working out to support answer	2
• One error made	1

Sample answer

$$\begin{aligned} I &= 12000 \times 0.024 \times \frac{8}{12} \\ &= \$192 \end{aligned}$$

$$\therefore 12\,000 + 192 = \$12\,192$$

19 (a) (ii)

Criteria	Marks
• Correct solution with full working out to support answer	3
• One error made	2
• Two errors made	1

Sample answer

$$365.75 = 12\,192 \times r \times 1$$

$$r = \frac{365.75}{12\,192} \times 100$$

$$r = 2.9999 \dots$$

$$= 3\%$$

19 (b) (i)

Criteria	Mark
• Correct solution	1

Sample answer

\$679.00

19 (b) (ii)

Criteria	Marks
• Correct solution with full working out to support answer	3
• One error made	2
• Two errors made	1

Sample answer

Maximum youth allowance possible for Deanne is \$530.40.

Annual reduction:

$$49\,302 - 45\,000 = 4\,302$$

$$4\,302 \times 0.17 = 731.34$$

$$731.34 \div 26 \text{ fortnights} = \$28.13$$

$$\therefore 530.40 - 28.13 = \$502.27$$

Question 20

20 (a) (i)

Criteria	Mark
• Correct solution	1

Sample answer

2016

20 (a) (ii)

Criteria	Mark
• Correct solution	1

Sample answer

Answers may vary.

Possible answer could include: Covid restrictions resulted in the Royal Easter Show being cancelled and therefore no one could attend.

20 (b) (i)

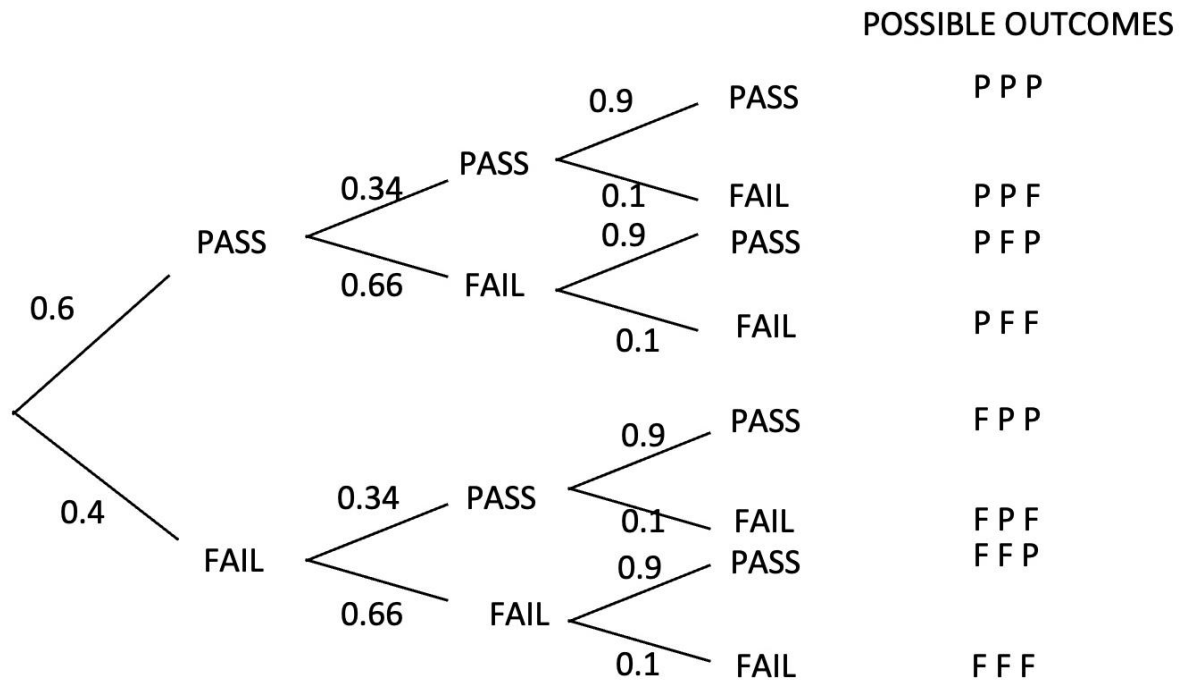
Criteria	Mark
• Correct solution	1

Sample answer $2 \times 2 \times 2 = 8$ possible outcomes.

20 (b) (ii)

Criteria	Marks
Must correctly construct tree diagram, including all three events with possibilities on each branch AND possible outcomes listed	3
Mostly correct working	2
Some correct working	1

Sample answer



20 (b) (iii)

Criteria	Marks
• Correct solution with working to support answer	2
• One error made	1

Sample answer

$$\begin{aligned}P(\text{passing at least 2 sections}) &= 1 - P(FFF) \\&= 1 - (0.4 \times 0.66 \times 0.1) \\&= 0.9736\end{aligned}$$

20 (c)

Criteria	Marks
• Correct solution with working to support answer	2
• One error made	1

Sample answer

$$Q_1 = 4$$

$$Q_3 = 7$$

$$\begin{aligned}IQR &= 7 - 4 \\&= 3\end{aligned}$$

$$7 + 1.5 \times 3 = 11.5$$

$\therefore$ Yes, there is an outlier, 12.

2022 Year 11 Mathematics Standard Mapping Grid

Section I

Question	Marks	Content	Syllabus Outcomes
1	1	Applications of Measurement	MS-M1.1
2	1	Working with Time	MS-M2
3	1	Formulae and Equations	MS-A1
4	1	Money Matters	MS-F1.1
5	1	Applications of Measurement	MS-M1.1
6	1	Linear Relationships	MS-A2
7	1	Working with Time	MS-M2
8	1	Applications of Measurement	MS-M1.2
9	1	Formulae and Equations	MS-A1
10	1	Formulae and Equations	MS-A2
11	1	Applications of Measurement	MS-M1.3
12	1	Money Matters	MS-F1.1
13	1	Money Matters	MS-F1.3
14	1	Applications of Measurement	MS-M1.2
15	1	Data Analysis	MS-S1.1

Section II

Question	Marks	Content	Syllabus Outcomes
16 (a)	2	Formulae and Equations	MS-A1
16 (b)	6	Formulae and Equations	MS-A1
16 (c)	6	Linear Relationships	MS-A2
17 (a)	6	Applications of Measurement	MS-M1
17 (b)	6	Applications of Measurement	MS-M1
17 (c)	3	Applications of Measurement	MS-M1
17 (d)	3	Applications of Measurement	MS-M1
17 (e)	4	Applications of Measurement	MS-M1, MS-F1
18 (a)	2	Working with Time	MS-M2
18 (b)	4	Working with Time	MS-M2
18 (c)	4	Working with Time	MS-M2
19 (a)	5	Money Matters	MS-F1
19 (b)	4	Money Matters	MS-F1
20 (a)	2	Data Analysis	MS-S1
20 (b)	4	Relative Frequency and Probability	MS-S2
20 (c)	2	Data Analysis	MS-S1